DEPENDENCY AWARE PQC MIGRATION FRAMEWORK FOR DIGITAL BANKING

End-to-end system architecture | Graph-aware migration planning and risk-bound banking transactions

1 ASSET INVENTORY & DEPENDENCY MODEL

Synthetic data and service-weighted risk index

Banking inventory

2,100 assets | 60 services
2,862 dependency records

Assets / services / dependencies CSVs

Load & validate

Schema, tiers, IDs, ranges, dependency endpoints

src/data_loader.py

Build directed graph

Internal / third-party / sub-tier
2,856 unique weighted edges

src/graph_builder.py

Evaluate ecosystem risk

Direct risk + strongest dependency paths + service-weighted ER

src/risk_model.py | src/propagation.py

risk context

2 RISK-BASED MIGRATION PLANNER

Budgeted counterfactual planning, not live deployment

Candidate risk impact

Temporarily set vulnerability to 0
Recompute modeled ΔER

src/remediation.py

Priority calculation

Choose largest positive ΔER / cost within remaining budget

src/prioritization.py

Dynamic migration plan

Select asset; update model; rerank
Budget ceilings: 2.5 / 5.0 / 7.5

experiments/final_evaluation.py

Planning outputs

Ranked asset list, migration trace
New modeled ER; comparison CSVs

results/tables | results/figures

policy refresh

3 TRANSACTION-TIME PQC ENFORCEMENT

ML-KEM / ML-DSA with simulated supplier evidence

Risk-bound policy

Standard: ML-KEM-768 / ML-DSA-65
Elevated: ML-KEM-1024 / ML-DSA-87

pqc_layer/darp_pqc.py

Banking gateway + RBAC

Service policy, actor authorization
Epoch & policy-digest binding

pqc_layer/banking_gateway.py

Cryptographic protection

OpenSSL ML-KEM, ML-DSA
HKDF-SHA256 + AES-256-GCM

pqc_layer/crypto_backend.py

Verify and accept

Signature + binding + replay checks
Reject missing / invalid tier evidence

pqc_layer/multitier_assurance.py

Cross-tier evidence — Simulated third-party + sub-tier signed acknowledgements — **enforced at gateway**

4 EVALUATION & ARTIFACTS

Planning and cryptographic tests reported separately

Migration / graph evaluation

Baselines, adapted QARS, budget sweeps, ablation, scalability

Security / performance checks

PQC operations, policy changes, tier-gate tests, latency, envelope size

Interpretation: migration reduces a synthetic modeled-risk index; supplier evidence is issued by simulated providers, not independently attested real vendors.